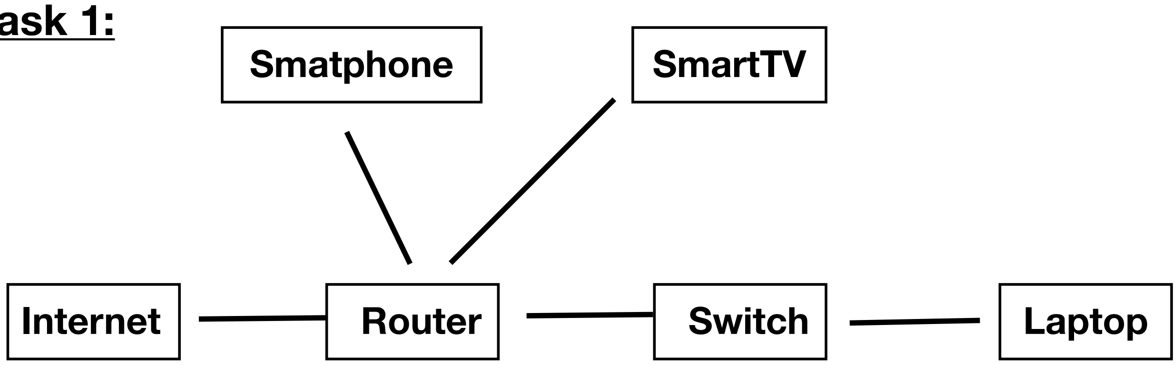
Module 1: Network + Network Fundamentals and building networks

Network Devices

Task 1:



Home network diagram: Internet enters through the router, which broadcasts Wi-Fi directly to the **smartphone** and **smart TV**, and also connects via a wired Ethernet cable to a **switch**, which then wires up the **laptop** for a stable, high-speed connection.

Task 2:

	Use case
Router	Your home router is what lets you stream on YouTube or Netflix — it fetches data from the internet and directs it to whichever device is watching
Switch	Used in a college computer lab or LAN gaming setup — connects many PCs together for fast, low-latency local multiplayer or file sharing without wasting bandwidth
Hub	Rarely used today, but historically found in very small, cheap office setups where a handful of computers just needed basic connectivity.

	Use case
Access Point	Used in large houses, offices, or colleges to give good Wi-Fi signal in rooms far from the main router — e.g., getting solid Instagram/WhatsApp connectivity in a back bedroom or hostel corridor

Task 3:

How data travels: ordering food on Zomato

You open the Zomato app on your phone and tap "place order." Your phone sends this request over Wi-Fi to the nearest **access point** (or directly to your **router** if you're close to it), which picks up the wireless signal and converts it into a form the wired network understands. The **router** then looks at where this data needs to go — since it's destined for Zomato's servers on the internet, not another device at home — and forwards it out through your ISP connection. If your home network uses a **switch** to connect other wired devices (like a smart TV or PC), the switch isn't directly involved in this particular request, but it keeps those devices connected on the same LAN so they could all reach the router simultaneously. The router keeps track of your specific request (using NAT) so that when Zomato's servers respond with your order confirmation, it knows to send that response back specifically to your phone rather than any other device on the network.

Task 4:

Gaming PC far from the router, only has an Ethernet port

Two good options:

1. **Powerline adapter** — Plug one adapter into a wall socket near your router (connected to the router via a short Ethernet cable), and plug the second adapter into a wall socket near your gaming PC (connected to the PC via Ethernet). The adapters send network data through your home's existing electrical wiring, effectively turning your power lines into a network cable — no need to run a long Ethernet cable across the house.

2. **Wi-Fi extender/bridge (with Ethernet output)** — Place a Wi-Fi extender or a wireless bridge near your gaming PC. It connects wirelessly back to your router's Wi-Fi signal, then provides an Ethernet port that plugs directly into your PC — giving you a stable wired connection at the PC's end while only needing Wi-Fi range in between, no new cabling required.